

CAMBRIDGE LEARNING GROUP
MARKED SCRIPT REPORT

Secondary 3 Express — End-of-Year Examination — Additional Mathematics 4049

Candidate	Class	Index	Paper
Chloe Yong	3S4	19	Paper 1 (2 h 15 min)

56 / 90

62.2% — grade band B4

Marks by question

Q	Topic	Mark	Lost
1	Surds — area of a triangle	2 / 3	-1
2	Surds — rationalising	4 / 4	—
3	Compound angles	1 / 4	-3
4	Completing the square / discriminant	4 / 4	—
5	Discriminant — line meets curve	3 / 5	-2
6	Logarithms — change of base	4 / 5	-1
7	Partial fractions	5 / 6	-1
8	Coordinate geometry	5 / 7	-2
9	Trigonometric graphs	7 / 7	—
10	Exponential growth	5 / 7	-2
11	Binomial theorem	1 / 8	-7
12	Polynomials	5 / 10	-5
13	Linear law	7 / 8	-1
14	Trigonometric identities	3 / 12	-9
	TOTAL	56 / 90	-34

Where the 34 marks went

Left blank or abandoned — 18 marks

Q3 (3), Q5 (2), Q11 (7), Q12(b) (3), Q14(c) (3). Every one of these was a question she had already started or could have started — the marks were available, not out of reach.

Method / concept error — 9 marks

Q8(ii) used midpoint of AB = midpoint of DC instead of the diagonals (2); Q14(a) combined the two fractions without squaring the numerators (5); Q10(b) tripled \$50 000 instead of the \$45 000 investment (1); Q7(i) stated the correct form but gave no reason (1).

Careless slip — 3 marks

Q1 expanded $\sqrt{3}(\sqrt{3}+2)$ wrongly (1); Q12(a) wrote $(-2)^2$ where the working needed $(-3)^2$ (2). Both were single characters that cost the whole answer.

Right work, wrong form — 4 marks

Q6 rejected the root for the wrong reason (1); Q10(b) gave $t = 12$ years instead of the year 2025 (1); Q14(b) answered in degrees when the range was in radians (1); Q13 misread two points off her own line (1).

The headline: 18 of the 34 lost marks were never contested. Chloe's method is sound far more often than her score suggests — she scored full marks on five questions and dropped only one mark on three more. What separates 56 from the high 70s is finishing the questions she starts and checking the form the answer is asked for.

Question by question

Question 1 Surds — area of a triangle

2 / 3

She wrote $\frac{1}{4}(7\sqrt{3}+9)=\frac{1}{2}(DF)(\sqrt{3}+2)\sin 60^\circ$, then expanded $\sqrt{3}(\sqrt{3}+2)$ as $\sqrt{3}+2\sqrt{3}+4$ and reached $DF=\frac{27-\sqrt{3}}{44}$.

Awarded M1 M1 Lost A1

Why the marks went The area formula was set up correctly and the rationalising was carried out correctly on her own expression, so both method marks stand. But $\sqrt{3}(\sqrt{3}+2)=3+2\sqrt{3}$, not $\sqrt{3}+2\sqrt{3}+4$ — $\sqrt{3}\times\sqrt{3}=3$. One slip, and the answer mark is gone.

Correct $DF=5-\sqrt{3}$ cm

Question 2 Surds — rationalising the denominator

4 / 4

She wrote (a) $\frac{7(3+\sqrt{2})}{9-2}=3+\sqrt{2}$. (b) $\frac{(5+4\sqrt{3})(2-\sqrt{3})}{4-3}=-2+3\sqrt{3}$.

Awarded M1 A1 M1 A1

Why the marks went Clean, complete and correctly simplified in both parts. Nothing to add.

Question 3 Compound angles

1 / 4

She wrote A CAST diagram with a 3–4–5 triangle placed in the third quadrant, labelled B . No further working and no answer.

Awarded B1 Lost B1 M1 A1

Why the marks went The diagram shows she had worked out that B lies in quadrant 3, which is the whole point of the question — that earns the first B1, and generously so, since nothing was written about A . From there it was three routine lines: $\cos A=\frac{-2}{\sqrt{5}}$, $\sin B=\frac{-3}{5}$, $\cos B=\frac{-4}{5}$, then the formula. Three marks for work she had already half-done in her head.

Correct $\cos(A-B)=\frac{11}{5\sqrt{5}}=\frac{11\sqrt{5}}{25}$, so $p=\frac{11}{25}$

Question 4 Completing the square and the discriminant

4 / 4

She wrote (i) $y=-2(x+3)^2+23$. (ii) $b^2-4ac\leq 0$ giving $184-8k\leq 0$, so $k\geq 23$.

Awarded M1 A1 M1 A1

Why the marks went Both parts fully correct, and she included $k=23$ rather than writing a strict inequality — that is the mark most candidates lose here.

Question 5 Discriminant — line meets curve

3 / 5

She wrote $nx^2+2x+(1-n)=0$, then $4-4n+4n^2>0$. Stopped there.

Awarded M1 A1 M1 Lost M1 A1

Why the marks went Everything written is right. She simply stopped one step early. The question says “show”, so it needs the reason the expression is always positive: $n^2 - n + 1 = \left(n - \frac{1}{2}\right)^2 + \frac{3}{4} \geq \frac{3}{4} > 0$, therefore two distinct points. Two marks for two lines.

Question 6 Logarithms — change of base

4 / 5

She wrote $\log_3 \sqrt{9} = 1$, changed base correctly, reached $(x-4)(x-1) = 0$, then “reject $x=1$ since $x > 0$ ”.

Awarded B1 M1 M1 A1 Lost A1

Why the marks went The mathematics is right and $x=4$ is the right answer, but the reason for rejecting $x=1$ is wrong: $1 > 0$, so “ $x > 0$ ” does not reject anything. The binding condition is $x-3 > 0$, i.e. $x > 3$. The final mark is specifically for that justification.

Question 7 Partial fractions

5 / 6

She wrote (i) “ $\frac{A}{x+1} + \frac{Bx+C}{x^2+4}$ should be how it is expressed.” (ii) Full working to $A=2, B=1, C=2$.

Awarded B1 M1 M1 A1 A1 Lost B1

Why the marks went Part (ii) is flawless. Part (i) asked her to

explain — stating the correct form is not a reason. The reason is that x^2+4 is irreducible, so it needs a numerator one degree lower, $Bx+C$; a constant leaves two unknowns to match three coefficients.

Correct $\frac{2}{x+1} + \frac{x+2}{x^2+4}$

Question 8 Coordinate geometry

5 / 7

She wrote (i) $m_{AC} = -2, m_{BC} = \frac{1}{2}$, product -1 . (ii) midpoint of $AB =$ midpoint of DC , giving $D(-2, 1)$.

(iii) Shoelace on her own points $\Rightarrow 10$ units².

Awarded M1 M1 A1 M1 A1 Lost M1 A1

Why the marks went Part (i) is correct. Part (ii) uses a property that is not true: AB and DC are opposite sides of the parallelogram, and their midpoints are not equal. The property needed is that the diagonals bisect each other, so midpoint $AC =$ midpoint BD , giving $D(6, 5)$. Part (iii) keeps both marks on follow-through — the shoelace method was applied correctly, and it happens to give 10, the right answer.

Question 9 Trigonometric graphs

7 / 7

She wrote $a=7, b=-3$; amplitude 7, period π ; and a sketch of $y=7 \cos 2x-3$ with two full cycles, maxima 4 at $x=0, \pi, 2\pi$ and minima -10 at $\frac{\pi}{2}, \frac{3\pi}{2}$, with $y=-3$ dashed in.

Awarded M1 A1 A1 B1 B1 B1 B1

Why the marks went The best question on the paper. The sketch is textbook — correct starting point, correct number of cycles, both turning values marked, and the centre line drawn in. She gave the amplitude as 7 and not 14, which is the classic trap here.

Question 10 Exponential growth

5 / 7

She wrote (a) $k = \ln \frac{50\,000}{45\,000} = 0.10536$. (b) $150\,000 = 45\,000 e^{0.10536t}$, $t = 11.43 \approx 12$.

Awarded M1 M1 A1 M1 M1 **Lost M1 A1**

Why the marks went Part (a) is perfect. In part (b) “tripled” means three times the investment:

$3 \times 45\,000 = 135\,000$ dollars — she used three times the \$50 000 valuation of 2016. Her algebra after that is correct, so the two method marks follow through. The final mark also needs the

year, not the duration: $t = 10.43$ years after the start of 2015 falls in 2025.

Question 11 Binomial theorem

1 / 8

She wrote (a) The first three terms of the expansion written out, no general term and no answer. (b) Only $T_{r+1} = \binom{n}{r} a^{n-r} b^r$ and “ $x^7 : x^9 = 4 : 7$ ”.

Awarded M1 **Lost M1 A1 M1 M1 M1 M1 A1**

Why the marks went This is the single biggest loss on the paper and it is a technique gap, not carelessness.

Writing out terms one by one cannot find the term independent of x — the general term is needed so the power of x can be set to zero. With the $3x^3$ in front, the power is $15 - 3r$, so $r = 5$ and the term is -297 . In (b) the two coefficients are $\binom{15}{7} 2^8 m^7$ and $\binom{15}{9} 2^6 m^9$; their ratio gives $m^2 = 9$, so $m = \pm 3$. She quoted the right formula and then did not use it.

Question 12 Polynomials

5 / 10

She wrote (a) $f(-3) = 0$ written as $(-3)^3 + (-2)^2 + a(-3) + 2b = 0$, leading to $a = \frac{-57}{13}$, $b = \frac{64}{13}$. (b)

Factorised to $(x-1)(x^2-7)$ and stopped.

Awarded M1 M1 A1 M1 A1 **Lost A1 A1 M1 A1 A1**

Why the marks went In (a) she wrote $(-2)^2$ where the substitution needs $(-3)^2$. One character: the equation became $-3a + 2b = 23$ instead of $-3a + 2b = 18$, and the fractional answers that followed should have been a warning sign — a and b were meant to come out as -4 and 3 . Her second equation and her method were both right.

In (b) the factorisation is correct and worth two marks, but the question says “solve”: $x = 1$, $x = \sqrt{7}$, $x = -\sqrt{7}$. Three more marks sat one line away.

Question 13 Linear law

7 / 8

She wrote $\lg L = k \lg d + \lg A$ with Y, m, X, c labelled; both $\lg$ rows computed; points plotted and a line drawn; then $k = 0.406$ and $A = 3.27$.

Awarded B1 B1 M1 A1 M1 M1 A1 **Lost A1**

Why the marks went Strong work — the linearisation, the table and the plot are all correct, and the intercept method is right. She then read $(1.2, 0.95)$ and $(2.8, 1.6)$ off her own line; the line actually passes nearer $(1.2, 0.99)$ and $(2.8, 1.57)$, which gives $k \approx 0.365$ rather than 0.406 . Read the two points as far apart as the line allows and take them where the line crosses a grid intersection.

Question 14 Trigonometric identities

3 / 12

She wrote (a) $\text{LHS} = \frac{\cos \theta - (1 - \sin \theta)}{\cos \theta (1 - \sin \theta)}$, then stopped. (b) $2 \tan \theta = \cot \theta$ through to $35.3^\circ, 144.7^\circ, 215.3^\circ, 324.7^\circ$. (c) Blank.

Awarded M1 M1 A1 Lost M1 M1 M1 M1 A1 A1 M1 M1 A1

Why the marks went Part (a) fails at the first line. Putting $\frac{p}{q} - \frac{r}{s}$ over a common denominator gives $\frac{ps - rq}{qs}$ — the numerators must be multiplied by the other denominator. The correct first line is $\frac{\cos^2 \theta - (1 - \sin \theta)^2}{(1 - \sin \theta) \cos \theta}$, and from there the numerator factorises to $2 \sin \theta (1 - \sin \theta)$ and the identity falls out in two more lines.

Part (b) is the encouraging part: she used the given result, reached $\tan^2 \theta = \frac{1}{2}$ and found all four angles correctly — but in degrees, when the range $0 \leq \theta \leq 2\pi$ is in radians. The answers are $0.615, 2.53, 3.76, 5.67$ rad. One conversion away from full marks.

Part (c) was left blank. It is three marks and four lines: the identity turns it into $kt^2 - 2t + k = 0$ with $t = \tan \theta$, and no solutions means discriminant < 0 , giving $k < -1$ or $k > 1$.

What to work on, in order

1. Finish the question you started

18 of the 34 lost marks were on work she had already begun. Q5 needed two lines, Q12(b) three, Q14(c) four. A hard rule for the next paper: before moving on, re-read the command word — show, solve, find the year, give the set — and check the answer is in that form.

2. Binomial theorem — the general term

The weakest topic by a distance, 1 out of 8. She knows the formula but does not use it as a tool. Drill only this: write the general term, collect the power of x , set it to the power required, solve for r . Ten questions of that one move and the topic is fixed.

3. Combining algebraic fractions

The Q14(a) error cost 5 marks and it is arithmetic, not trigonometry: $\frac{p}{q} - \frac{r}{s} = \frac{ps - rq}{qs}$. Worth checking this is secure outside a trig context first, because it will keep reappearing.

4. Check the substitution, then check the answer looks sensible

Q1 and Q12(a) were single-character slips — $\sqrt{3} \times \sqrt{3}$ and $(-3)^2$ — worth 3 marks. In Q12 the fractional $a = \frac{-57}{13}$ was itself the signal that something was wrong, since the question implies whole numbers. Teach her to treat an ugly answer as a prompt to re-check.

5. Radians and degrees

Q14(b) had all four angles right and scored 3 of 4 because the range was in radians. A quick habit: look at the range before starting, and set the calculator to match.

Overall. 56 / 90 understates her. Full marks on Q2, Q4 and Q9, and one mark off on Q6, Q7 and Q13, show the core is in place across surds, quadratics, logarithms, partial fractions, graphs and linear law. Two topics need real teaching — binomial expansion and proving identities — and the rest is exam discipline. Both are fixable before the next paper.